

A negative answer to a question of Kierstead, Kostochka and Xiang

on strongly equitable list coloring of forests

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Abstract

A forest T on n vertices is equitably k -colorable ($k \geq 3$) if and only if every vertex lies in an independent set of size at least $\lfloor n/k \rfloor$ (Chen–Lih; Miyata–Tokunaga–Kaneko). Kierstead, Kostochka and Xiang asked whether the same condition characterizes strongly equitable k -choosability of forests. We show that it does not: for every $k \geq 2$ there is a forest, and even a tree, on $2k^3 + k$ vertices that satisfies the independence condition (indeed is equitably k -colorable) but is not equitably k -choosable even in the weaker sense of Kostochka, Pelsmajer and West. The construction sharpens an example of Kaul, Mudrock and Wagstrom, and it generalizes to a family of necessary conditions for strongly equitable choosability of forests.

1 Introduction

All graphs are finite and simple. For a graph G and a vertex v , let $\alpha_v(G)$ denote the maximum size of an independent set of G containing v , and let $N(v)$, $N[v] = N(v) \cup \{v\}$ and $d(v) = |N(v)|$ be the neighborhood, closed neighborhood and degree of v . We write $[k] = \{1, \dots, k\}$.

A proper vertex coloring is *equitable* if any two color classes differ in size by at most one. In an equitable k -coloring of an n -vertex graph every class has at least $\lfloor n/k \rfloor$ vertices, so the class of v witnesses $\alpha_v(G) \geq \lfloor n/k \rfloor$. For forests this obvious necessary condition is also sufficient.

Theorem 1 (Chen–Lih [2]; Miyata–Tokunaga–Kaneko [8]; see Chang [1]). *Let $k \geq 3$. A forest T on n vertices is equitably k -colorable if and only if $\alpha_v(T) \geq \lfloor n/k \rfloor$ for every vertex v .*

We call the condition in Theorem 1 the *independence condition*.

List versions

A k -list assignment for G is a map L from $V(G)$ to sets of colors with $|L(v)| = k$ for all v , and an L -coloring is a proper coloring f with $f(v) \in L(v)$ for all v . Kostochka, Pelsmajer and West [7] call an L -coloring *equitable* if every color class has at most $\lceil |G|/k \rceil$ vertices, and G *equitably k -choosable* if it has an equitable L -coloring for every k -list assignment L . Note that an equitable L -coloring may use more than k colors and may have very small classes; only an upper bound on class sizes is imposed.

Kierstead, Kostochka and Xiang [4, 5] introduced a stronger notion. Write $|G| = kq + r$ with $q \in \mathbb{N}$ and $1 \leq r \leq k$, and set $|G| \bmod^* k := r$ (so $|G| \bmod^* k = k$ when k divides $|G|$). A color class is *full* if it has exactly $\lceil |G|/k \rceil$ vertices. An L -coloring is *strongly equitable* (SE) if no class

has more than $\lceil |G|/k \rceil$ vertices and at most $|G| \bmod^* k$ classes are full; G is SE k -choosable if it has an SE L -coloring for every k -list assignment L . Two features of this definition matter here.

- Every SE L -coloring is an equitable L -coloring in the sense of [7]; hence SE k -choosable graphs are equitably k -choosable.
- If all lists equal $[k]$, then the SE L -colorings are exactly the equitable k -colorings: with $r = |G| \bmod^* k$ there are r classes of size $q+1$ and $k-r$ of size q . Hence SE k -choosable graphs are equitably k -colorable, and in particular SE k -choosable forests satisfy the independence condition.

In their survey [6, Question 13], Kierstead, Kostochka and Xiang asked whether Theorem 1 extends to this setting.

Question 2 ([6]). Is it true that for each $k \geq 3$, an n -vertex forest T is SE k -choosable if and only if $\alpha_v(T) \geq \lfloor n/k \rfloor$ for every vertex v ?

By the second feature above, the “only if” direction holds. We show that the “if” direction fails for every k , even for trees, and even for the weaker notion of [7].

Theorem 3. *Let $k \geq 2$, $q = 2k^2 + 1$ and $n = kq = 2k^3 + k$. There is a forest F_k and a tree T_k , each on n vertices, such that*

- (i) F_k and T_k have equitable k -colorings; in particular $\alpha_v \geq q = \lfloor n/k \rfloor$ for every vertex v ;
- (ii) F_k and T_k are not equitably k -choosable in the sense of [7], and hence not SE k -choosable.

Consequently the answer to Question 2 is negative for every $k \geq 3$.

The forest F_k is a disjoint union of two stars. Kaul, Mudrock and Wagstrom [3, Proposition 22] proved, for a different purpose, that $K_{1,(k-1)(k^3-k+2)} + K_{1,k^3}$ is not equitably k -choosable. That forest also satisfies the independence condition, so it already answers Question 2 in the negative; the connection does not seem to have been noticed. Our F_k is the smallest member of the same family of examples (Remark 4). In Section 3 the argument is generalized to a family of necessary conditions for SE k -choosability of forests, and Section 4 collects open questions.

2 The construction

Throughout this section $k \geq 2$, $q = 2k^2 + 1$ and $n = kq$. Since k divides n , we have $\lfloor n/k \rfloor = \lceil n/k \rceil = q$, and an L -coloring is equitable in the sense of [7] exactly when every color is used at most q times.

The forest

Let F_k be the disjoint union of two stars: one with center u_0 and leaf set A , where $|A| = (k-1)q - 1$, and one with center w_0 and leaf set B , where $|B| = 2k^2$. Then

$$|F_k| = 2 + ((k-1)q - 1) + 2k^2 = (k-1)q + q = kq = n.$$

Index the leaves of w_0 by triples, $B = \{b_{c,d,t} : c \in [k], d \in \{k+1, \dots, 2k\}, t \in \{1, 2\}\}$.

Equitable colorability

The set $\{u_0\} \cup B$ is independent and has $1 + 2k^2 = q$ vertices; the set $\{w_0\} \cup A$ is independent and has $(k - 1)q$ vertices. Using $\{u_0\} \cup B$ as one class and splitting $\{w_0\} \cup A$ into $k - 1$ classes of size q gives an equitable k -coloring of F_k . This proves (i) of Theorem 3 for F_k .

The list assignment

Define a k -list assignment L by

$$L(u_0) = L(a) = [k] \quad (a \in A), \quad L(w_0) = \{k + 1, \dots, 2k\}, \quad L(b_{c,d,t}) = ([k] \setminus \{c\}) \cup \{d\}.$$

No equitable L -coloring

Suppose f is an L -coloring of F_k in which every color is used at most q times. Let $c = f(u_0) \in [k]$ and $d = f(w_0) \in \{k + 1, \dots, 2k\}$. Consider the set

$$S = A \cup \{b_{c,d,1}, b_{c,d,2}\}, \quad |S| = (k - 1)q + 1.$$

Every $a \in A$ is adjacent to u_0 and has list $[k]$, so $f(a) \in [k] \setminus \{c\}$. Each of $b_{c,d,1}, b_{c,d,2}$ is adjacent to w_0 , so it does not receive d , and its remaining colors are $[k] \setminus \{c\}$. Thus f maps the $(k - 1)q + 1$ vertices of S into the $k - 1$ colors of $[k] \setminus \{c\}$, and by the pigeonhole principle some color is used more than q times, a contradiction. Hence F_k is not equitably k -choosable, and, since every SE L -coloring is an equitable L -coloring, not SE k -choosable. This proves (ii) for F_k .

The tree

Fix $a_0 \in A$ and $b_0 \in B$ and let $T_k = F_k + a_0b_0$. Adding an edge cannot create an L -coloring, so T_k is not equitably k -choosable. For (i), $\{u_0\} \cup B$ and $\{w_0\} \cup A$ are still independent in T_k (neither contains both a_0 and b_0), and the equitable k -coloring above places a_0 and b_0 in different classes, so it is a proper equitable k -coloring of T_k . This completes the proof of Theorem 3. $\square$

Remark 4. In the construction, the leaves of u_0 fill the $k - 1$ colors $[k] \setminus \{c\}$ up to a *slack* of exactly one vertex, and for each of the k^2 possible pairs $(f(u_0), f(w_0))$ two leaves of w_0 compete for that single slot. The example of [3, Proposition 22] is the same mechanism with slack $k - 1$ and k competing leaves per pair, which costs $n = k(k^3 - k + 3)$ vertices instead of $2k^3 + k$ (81 instead of 57 for $k = 3$). Slack zero is impossible: if $|A| = (k - 1)q$ then $|B| = q - 2$ and $\alpha_{u_0} = q - 1$, which violates the independence condition. This is precisely the obstruction that makes Theorem 1 true, and the list version escapes it by one vertex.

3 A general obstruction

The proof of Theorem 3 only used a count of vertices forced into $k - 1$ colors. We record the general form of this count.

Let T be a forest on n vertices and write $n = kq + r'$ with $0 \leq r' \leq k - 1$, so that $q = \lfloor n/k \rfloor$. In an SE L -coloring, any $k - 1$ colors together carry at most $n - q$ vertices: if $r' = 0$ every class has at most q vertices, and if $r' \geq 1$ every class has at most $q + 1$ vertices with at most r' classes of size $q + 1$, so the bound is $(k - 1)q + r' = n - q$ in both cases. In an equitable L -coloring in the sense of [7] the corresponding bound is $(k - 1)\lceil n/k \rceil$.

Proposition 5. *Let T be a forest on n vertices, $q = \lfloor n/k \rfloor$, let $u \in V(T)$, let $W \subseteq V(T) \setminus \{u\}$, and let $Y_w \subseteq N(w) \setminus N[u]$ ($w \in W$) be pairwise disjoint. If*

$$d(u) - |W \cap N(u)| + \sum_{w \in W} \left\lfloor \frac{|Y_w|}{k^2} \right\rfloor > n - q, \quad (1)$$

then T is not SE k -choosable. If (1) holds with $n - q$ replaced by $(k - 1)\lceil n/k \rceil$, then T is not equitably k -choosable in the sense of [7].

Proof. Let $D = \{k + 1, \dots, 2k\}$. Define $L(u) = [k]$, $L(x) = [k]$ for $x \in N(u) \setminus W$, and $L(w) = D$ for $w \in W$. For each $w \in W$ split Y_w into k^2 parts $Y_{w,c,d}$ ($c \in [k]$, $d \in D$), each of size at least $\lfloor |Y_w|/k^2 \rfloor$, and give every $y \in Y_{w,c,d}$ the list $([k] \setminus \{c\}) \cup \{d\}$. All other lists are arbitrary k -sets. Let f be an SE L -coloring and put $c = f(u)$. Every $x \in N(u) \setminus W$ is colored from $[k] \setminus \{c\}$. For $w \in W$ let $d = f(w) \in D$; every $y \in Y_{w,c,d}$ is adjacent to w , cannot receive d , and is therefore colored from $[k] \setminus \{c\}$. The number of these vertices is at least the left-hand side of (1), which exceeds the capacity $n - q$ of $k - 1$ colors. The second statement is identical with the other capacity. $\square$

Theorem 3 is the case $W = \{w_0\}$, $Y_{w_0} = B$, $|B| = 2k^2$, $d(u_0) = (k - 1)q - 1$, where the left-hand side equals $(k - 1)q + 1 = n - q + 1$. With $W = \emptyset$, condition (1) reads $d(u) > n - q$; since $\alpha_u \leq n - d(u)$, this is already excluded by the independence condition. Thus Proposition 5 adds new information only when some vertex w has at least k^2 neighbors outside $N[u]$.

4 Concluding remarks and open questions

Verification

The forest F_k , the list assignment L , the independence condition and the non-existence of an SE L -coloring have been formalized for all $k \geq 2$ and machine-checked in Lean 4 with Mathlib (file `Q13lean/Counterexample.lean` in the accompanying repository). Independently, an exact flow-based program confirms Theorem 3 for $k = 3, 4$ and the example of [3] for $k = 3, 4$.

Small forests

An exhaustive computer search found no forest satisfying the independence condition and admitting a defeating k -list assignment in the following ranges, where the palette is the set of colors from which all lists are drawn.

k	number of vertices	palette size
3	≤ 12	4
3	≤ 10	5
4	≤ 11	5
5	≤ 10	6

Because a defeating assignment may need a larger palette (L above uses $2k$ colors), these searches do not yield a lower bound on the size of a smallest counterexample.

Question 6. For $k = 3$, what is the smallest number of vertices of a forest that satisfies the independence condition but is not SE 3-choosable? It is at most 57.

A corrected characterization?

The independence condition together with the family of conditions “(1) fails for all choices of u , W , (Y_w) ” is necessary for SE k -choosability of a forest. We do not know whether it is sufficient. A first test case is the class of disjoint unions of two stars, where even equitable k -choosability in the sense of [7] has not been characterized [3, Question 8].

Question 7. Is a forest T on n vertices SE k -choosable whenever $\alpha_v(T) \geq \lfloor n/k \rfloor$ for all v and (1) fails for all u , W and (Y_w) ? In particular, is T SE k -choosable whenever the independence condition holds and $\Delta(T) < k^2$?

For comparison, Kostochka, Pelsmajer and West [7] proved that every forest is equitably k -choosable (in the sense of [7]) when $k \geq 1 + \Delta/2$, and that this bound is tight (as quoted in [3]). No such degree bound can hold for SE k -choosability without the independence condition, since SE k -choosable graphs are equitably k -colorable; whether the independence condition suffices for SE k -choosability of forests with $\Delta \leq 2k - 2$ is a natural special case of the question above. By Proposition 5, the new list obstructions appear only at degree k^2 and beyond.

References

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